

Jesus Sanchez

Engineering Portfolio

Outline

1

About Me

2

Liquid Propulsion Lab

3

NASA JPL

4

Combustor Research

5

Skills

6

Coursework

7

Contact Information

About Me

My name is Jesus Sanchez, Aerospace Engineer from University of Southern California, with experience in propulsion and mechanical design.

I am motivated by the opportunity to solve complex challenges and contribute to advancements in technology that have a tangible impact on humanity's future. I am particularly inspired by purpose-driven work. Knowing that my work will directly contribute to groundbreaking technology motivates me to excel in every project I undertake.



Liquid Propulsion Lab (LPL)

(Sep 2024 - Present)

At USC's LPL, I worked on the design and testing of a regeneratively cooled 3 kN Jet-A/LOX rocket engine, with responsibilities spanning detailed design, thermal analysis, CFD, and FEA for the development of the injector, thrust chamber and nozzle.



Responsibilities

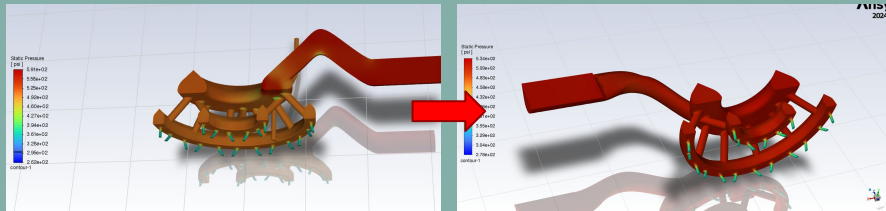
- CAD (Siemens NX)
- Injector, chamber, nozzle design
- CFD (injector, chamber and nozzle)
- MATLAB (thermal analysis, data analysis)
- Bolt analysis
- FEA



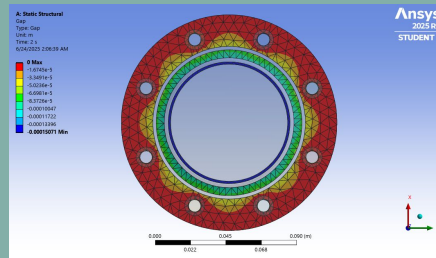
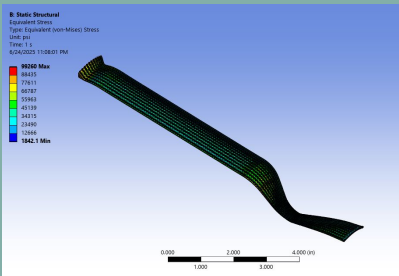
Liquid Propulsion Lab (LPL)

(Sep 2024 - Present)

CFD



FEA

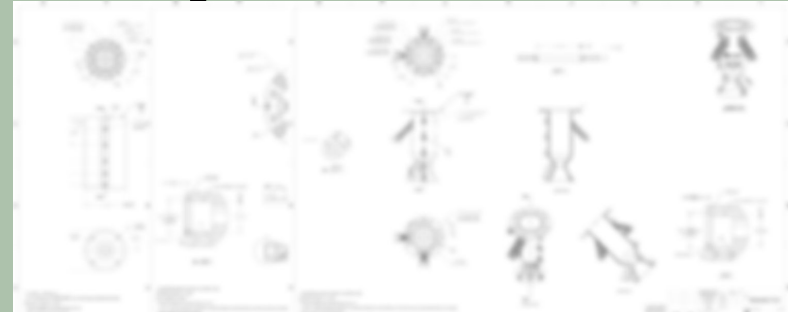


Bolt Analysis

Inputs		
Bolt		
Size, Material	5/16-18 Grade 8 Steel, Zinc Plated	
Nominal Diameter	Di	0.3125 [in]
		0.0250 [in]
Threads Per Inch (TPI)	TPI	18 [1/in]
Tensile Stress Area (Table B-2)	A	0.0524 [in ²]
Bolt Length	L	1.5 [in]
Bolt Thread Length	Lt	1 [in]
Bolt Head Width	D	0.5 [in]
Bolt Yield Strength	Yb	150000 [psi]
Effective Tensile Strength Percentage	%	0.75 [1]
Modulus of Elasticity	E	2.9E+07 [psi]
Number of Bolts	N	8 [1]
Inputs		
Torque Coefficient	K	0.2 [1]
Joint		
Grip Length	Lg	0.8 [in]
Nut Length	Ln	0.265625 [in]
Washers	W	0.3437 [in]
Top Member Elasticity	E1	2.90E+07 [psi]
Bottom Member Elasticity	E2	2.90E+07 [psi]
Alpha	a	30 [deg]
Top Member Thickness	LT	0.3 [in]
Bottom Member Thickness	LB	0.3 [in]
Loads		
Proof Load	Fp	1200 [psi]
Chamber Diameter	Dc	2.93 [in]
Sealing Forces		
O-ring Force	F1	310 [lbf]
Inner O-ring Force	F2	250 [lbf]

Stiffness		
Bolt Stiffness		
Length of unthreaded portion in grip	ld	0.5 [in]
Length of threaded portion in grip	lt	0.1 [in]
Area of unthreaded/major portion	At	0.0707 [in ²]
Area of threaded portion	At	0.0524 [in ²]
Faster Stiffness	kb	$k_b = \frac{A_b E}{L_b}$ 3.20E+06 [lbf/in]
Member Stiffness		
Inconel Stiffness, top	k1	$k = \frac{P}{\delta} = \frac{E A}{L}$ 2.38E+07 [lbf/in]
Inconel Stiffness, bottom	k2	2.38E+07 [lbf/in]
Total Member Stiffness	km	$\frac{1}{k_m} = \frac{1}{k_1} + \frac{1}{k_2}$ 1.19E-07 [lbf/in]
Joint Stiffness		
Total Joint Stiffness	C	$\frac{1}{C} = \frac{1}{k_b} + \frac{1}{k_m}$ 0.212 [1]
Loads		
Chamber Area	Ac	$\pi \cdot D_c^2 / 4$ 6.74 [in ²]
Chamber Force	Fc	$F_p \cdot A_c$ 8091.08 [lbf]
Total Separation Force	Fs	sum(F) 8651.08 [lbf]
Internal Separation Force per Bolt	Fpb	F_s / N 1081.38 [lbf]
Portion of Force Taken by the Bolt	Fb	$F_{pb} \cdot C$ 229.56 [lbf]
Portion of Force Taken by the Members	Fm	$(1-C) \cdot F_{pb}$ 851.82 [lbf]
Preload per Bolt	P	$Y_b \cdot A_b$ 5995.00 [lbf]
Total Preload	Pt	$P \cdot N$ 47160.00 [lbf]
Resultant Bolt Load	Fb_tot	$P + F_b$ 6124.56 [lbf]
Resultant Member Load	Fm_tot	$F_m \cdot P$ -5043.18 [lbf]
Bolt Stress Under Service	db	F_{b_tot} / A_b 11680.94 [psi]
Results		
Torque Required per Bolt	T	$P \cdot D_b \cdot K$ 30.70 [lb-ft]
FOS - Bolt Failure	FOS b	Y_b / db 1.28
FOS - Joint Separation	FOS s	P_t / F_s 6.92

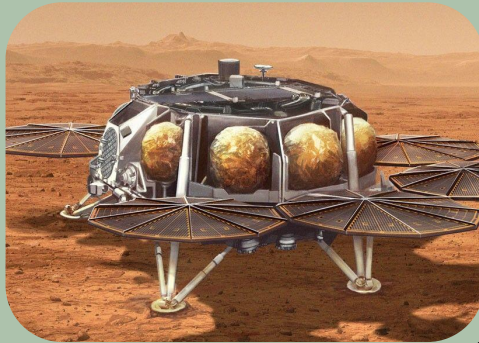
Drawings



NASA JPL

(May 2022 - Feb 2024)

During my internship, I worked as a Mechanical Engineer for the Electronic Packaging group. I began by designing ground support equipment (GSE) then transitioned to electromechanical systems for flight applications. Progressively, I led the design of an electromechanical unit for the Sample Retrieval Lander (SRL).



Responsibilities

- CAD (Siemens NX)
 - ◆ Design for manufacturing and assembly (DFM & DFA)
- Engineering Drawings
 - ◆ Part DWG
 - ◆ Assembly DWG
 - ◆ Interface DWG
- GD&T
- Written procedures for build and assembly
- Troubleshoot hardware assembly

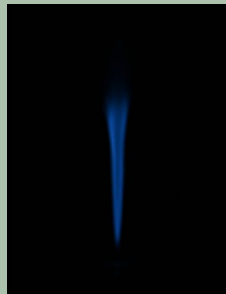
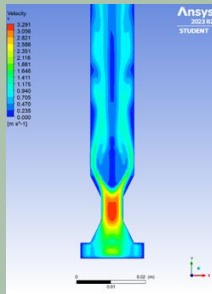
Combustor Research

(May 2023 - August 2024)

This research focused on the development of a novel swirl-injection combustor for gas turbine engines. The work aimed at investigating a new combustor and injection concept to improve air-fuel mixing and support stable operation with hydrogen fuel. I designed, simulated, and tested the combustor resulting in 60% hydrogen dilution level, surpassing current industry capabilities by 10%.

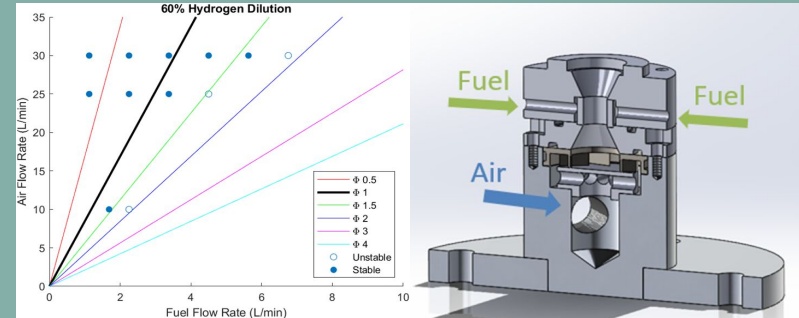
Conference paper:

<https://jesusfsanchez.github.io/ResearchPaper.pdf>



Responsibilities

- CAD (SOLIDWORKS)
- CFD (Ansys Fluent, OpenFOAM)
- MATLAB
- LabView
- GD&T



Skills

CAD, CFD, FEA

- Siemens NX
- SOLIDWORKS
(+Simulation)
- Ansys Fluent
- Ansys Mechanical
- OpenFOAM

Software

- MATLAB
- LabView
- Python
- Simulink
- Cantera
- Teamcenter

Technical

- GD&T
- Tolerance Analysis
- 3D Printing

Spacecraft Design

Spacecraft structures and mechanisms, space environment, orbital mechanics, attitude determination and control, power systems, thermal control

Rocket Propulsion

Liquid and solid propulsion, nozzles, propellants, feed system, flight performance, orbital mechanics & staging

Compressible Gas Dynamics

Shocks, expansions, Rayleigh flow, Fanno flow, unsteady wave motion, linearized potential theory, transonic, hypersonic flow, and CFD project

Machine Design

CAD, FEA, failure theory, bolt calculations, and design of mechanical components

Fundamentals and Applications of Combustion

Chemical thermodynamics, chemical kinetics, transport phenomena, non-premixed, premixed flames, Cantera and FDM project

Intro to Computational Fluid Dynamics

Fundamentals and general techniques used for numerical solution of fluid flow equations, including: numerical accuracy, stability, convergence, validation, and more

Coursework

Contact me

jesusfsa@usc.edu

(805)815-7654

<https://www.linkedin.com/in/jesus-sanchez1/>

<https://jesusfsanchez.github.io/>